EXPERIMENT-2

NAME-SEJAL

SECTION – 24AIT KRG 1-B

UID-24BAI70050

DATE-14/07/2026

SUBJECT-ADBMS

Q1.

(A)

A university administration requires an analytical performance report for each academic unit. Write a PostgreSQL solution to calculate the average marks scored by students grouped by their respective departments. The solution must join the student profile, course subjects, and grade marks tables, returning a breakdown sorted by the highest average marks down to the lowest.

```
SELECT D.DEPT_ID AS DEPARTMENT,
AVG(M.MARKS_SCORED) AS AVERAGE_MARKS
FROM DEPARTMENTS AS D
JOIN STUDENTS AS S
ON D.DEPT_ID=S.DEPT_ID
JOIN MARKS AS M
ON S.STUDENT_ID=M.STUDENT_ID
GROUP BY D.DEPT_NAME
ORDER BY AVERAGE_MARKS DESC;
```

Two legacy HR systems (A and B) have separate records of employee salaries. These records may overlap. Management wants to **merge these datasets** and identify **each unique employee (**by EmpID) along with their **lowest recorded salary** across both systems.

Objective

1. Combine two tables A and B.
2. Return each EmpID with their **lowest salary,** and the corresponding **Ename.**

```
SELECT EMPID,MIN(ENAME) AS ENAME,MIN(SALARY) AS SALARY
FROM
(
SELECT * FROM  A
UNION ALL
```

```
SELECT * FROM  B
) AS INTERMEDIAT_RESULT
GROUP BY EMPID
```

A multinational organization maintains separate records for employee information and department details. The management wants to identify the employees who receive the highest salary within their respective departments for an annual performance report.
If more than one employee earns the same highest salary in a department, all such employees should be included in the report. The final output should display the **Department Name**, **Employee Name**, and **Salary**, arranged in ascending order of department name.

Note: The employee and department information are maintained in separate normalized tables.

```
SELECT D.DEPT_NAME ,E.NAME,E.SALARY
FROM EMPLOYEE AS E
INNER JOIN DEPARTMENT AS D
ON E.DEPARTMENT_ID=D.ID
WHERE E.SALARY IN(
SELECT MAX(SALARY)
FROM EMPLOYEE AS E1
WHERE E1.DEPARTMENT_ID=E.DEPARTMENT_ID
AND
E1.SALARY NOT IN(
SELECT MAX(SALARY)
FROM EMPLOYEE AS E2
WHERE E2.DEPARTMENT_ID=E.DEPARTMENT_ID
)
)
```